MBEYA UNIVERSITY OF SCIENCE AND TECHNOLOGY

COLLAGE OF INFORMATION AND COMMUNICATION TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING.

COURSE NAME: APPLICATION DEVELOPMENT FOR MOBILE DEVICES.

COURSE CODE: CS8306

FACILITATOR: RUKOIJO A MSUYA

No	NAME	CA NUMBER
1.	JOSEPH JULIUS BALIJA	CA/BCS/25/13400
2.	CLEMENT M EDWARD	CA/BCS/25/14588
3.	EMMANUEL JOSEPH	CA/BCS/25/12372
4.	DEOGRATIUS MZAVA	CA/BCS/25/8763

SALES MANAGEMENT SYSTEM PROJECT REPORT

1. Introduction

Problem Description

Small and medium-sized businesses often struggle with managing product sales, tracking revenue, monitoring inventory, and generating analytical insights. Manual tracking methods are inefficient and prone to errors. A centralized digital system is needed to streamline operations.

Motivation

The motivation behind this project is to design and develop a web based Sales Management System that simplifies product and sales tracking, provides analytical insights, and enhances decision making.

Project Objectives

- Develop a user friendly sales tracking platform.
- Enable CRUD operations for products and sales.
- Provide analytics such as daily/monthly revenue and top selling products.

2. System Requirements

Functional Requirements

- Add, update, delete, and view products.
- Record and manage sales transactions.
- Generate sales analytics reports.
- Dashboard with visual charts.

Non-Functional Requirements

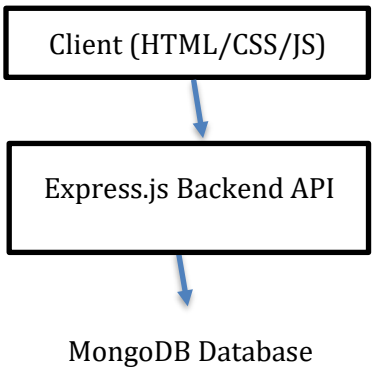
- System must be responsive and user friendly.
- Efficient database queries.
- Scalability for increasing data.

3. System Architecture

Overall System Design

The system follows a client server architecture where the frontend interacts with the backend through RESTful APIs. The backend processes requests and interacts with MongoDB.

Client-Server Architecture Diagram



Database Schema (ERD)
Entities:
1. Product (id, name, price, cost, stock, image)
2. Sale (id, product_id, quantity, totalPrice, profit, createdAt)

Security Measures Used

- Input validation and error handling.

4. Implementation Details

Technologies Used

- Frontend: HTML, CSS, JavaScript
- Backend: Node.js, Express.js
- Database: MongoDB, Mongoose

- Express error-handling middleware**.**

Frontend Implementation
The frontend is built using vanilla JavaScript with modular file structure (api.js, ui.js, state.js, products.js, sales.js, charts.js).

Backend Implementation
The backend uses Express.js to define RESTful API routes and controllers. Mongoose is used to define schemas and interact with MongoDB.

Database Design
MongoDB collections include Products and Sales. Relationships are handled using ObjectId references.

Challenges Faced
- Handling Express v5 routing changes.

- Managing asynchronous database operations.

- Ensuring correct static file serving.

Assumptions Made - Users have
stable internet access.

- The application will be deployed in a secure environment.

5. Security Considerations

Data Protection
Sensitive routes require authentication middleware.

Access Control Implementation
Role based access control can be implemented to restrict certain features.

6. Testing & Results Major

Features Tested:

- Product CRUD operations.
- Sales recording and profit calculation.
- Analytics dashboard charts.
- Incomplete Features (if any):
- Advanced role-based access control.
- Exporting analytics reports to PDF.

7. Future Work

Possible Improvements:

- Improve advanced reporting features.
- Add role based user management.
- Improve UI/UX design.

Scalability Considerations:

- Integrating with online payment systems like banks and mobile money
- Deploy using cloud infrastructure.

Video link: https://photos.app.goo.gl/TpwxM3SMU8ffPQjV8